

Disambiguating "Mating" and FAI Across the Trilogy

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2025

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026). It introduces no new axioms. Its sole contribution is to clarify the precise relationship between the term "mating" as used in Paper 2 and the term "FAI" as used in Paper 3, so that downstream work can adopt, extend, or argue against either term without conflating two architecturally distinct governance operations.

Abstract

The CKS trilogy uses two related but distinct terms for content-combination operations: "mating" in Paper 2, and "FAI" (Full Aspect Integration) in Paper 3. Paper 3 explicitly identifies FAI as inheriting the pattern structure of Paper 2's mating primitive at a larger governance authority scope. This inheritance relationship is precise but not identity: the two operations share the same three pattern variants, but they differ in participating entity scope (intra-Self vs. inter-Self), governance authority structure (one vs. joint), and output type (offspring within one Self vs. evolution feed delivered to each participating Self's home substrate). This note states the scope-specific meaning of each term, identifies the two properties that distinguish them, presents the one-to-one correspondence between Paper 2's mating variants and Paper 3's merge patterns, and provides a disambiguation rule practitioners can apply when reading across the trilogy.

1. Why disambiguation is needed

The word "mating" appears in Paper 2 as the name for a governed lifecycle operation that combines parental content to produce offspring. Paper 3 introduces FAI as the inter-Self coordination primitive and explicitly names Paper 2's mating as the architectural antecedent from which FAI inherits. This creates a risk in two directions.

The first risk is conflation: a reader of Paper 3 might interpret FAI as simply mating that happens to cross a Self boundary, missing the structural consequences of the different governance authority scope and the different output type. This misreading would lead to applying Paper 2's offspring-within-one-Self framing to FAI's evolution-feed-to-multiple-Selves mechanism — an error that would mischaracterize what FAI events produce and how those outputs reach participating Selves' evolution machinery.

The second risk is false novelty: a reader might argue that because FAI uses the same three pattern variants as Paper 2's mating, Paper 3 adds nothing at the coordination primitive level. This misreading would miss the structural-extension contribution: the same pattern at inter-Self scope under joint governance authority with evolution-feed outputs rather than offspring is not the same operation in architecturally relevant senses, even if the merge logic is inherited without modification.

This note resolves both risks by stating the scope-specific meanings precisely and identifying the exact properties that differ.

2. Mating as Paper 2's intra-Self primitive

Paper 2 (§6.3) establishes mating as one of three lifecycle primitives — alongside birth and death — that apply uniformly at every level of a CKS-governed Self: cell scope, aspect scope, and Self scope. The paper imports the combination-primitive concept from biology and evolutionary computation (where crossover has architectural status distinct from mutation) into AI Self architecture under governance.

In Paper 2's sense, a mating event has four defining properties:

Participating entities. Two or more parent entities from within *the same* Self. Mating is an intra-Self operation. The parents may be cells, aspects, or Selves at the scope at which the operation is applied, but they are all governed under the same governance authority — the Self whose lifecycle governance covers them.

Governance authority. One governance authority. The Self's governance — the human-held authority to inspect, modify, and override substrate content and orchestration rules — covers both parents and the offspring they produce. There is no cross-boundary authority negotiation.

Output. Offspring — new governed entities produced within the same Self. The offspring carry content from both parents, with the specific combination determined by whichever of the three pattern variants the orchestration substrate is configured to apply.

Three pattern variants:

Union: All content from both parents merges into the offspring; merge-time conflicts are preserved as first-class substrate state per Paper 1's conflict-preservation commitment, inherited directly.

Selective merge: Governance-directed curation determines which content crosses into the offspring; humans or LLMs operating under human direction pre-select before the combination executes.

Lineage-preserved union: All content merges, plus explicit cross-references to parent entities are written into the offspring so every element's provenance remains traceable.

"Mating" without qualification across the trilogy means Paper 2's intra-Self operation with these four properties.

3. FAI as Paper 3's inter-Self analog

Paper 3 (§5, §7) establishes FAI as the canonical operation over the shared substrate that participating Selves contribute aspects to. Paper 3's inheritance accounting (§7.6) explicitly identifies Paper 2's §6.3 intra-Self combination primitive as the architectural antecedent: FAI inherits the pattern structure from mating and extends it to the inter-Self scope.

FAI's four corresponding properties differ from mating's at two of the four:

Participating entities. Aspects from *different* Selves. FAI is an inter-Self operation. The contributing aspects carry their constituent cells' DNA-layer and action-layer content into a

temporary shared substrate that spans the governance perimeters of the participating Selves.

Governance authority. Joint authority across the governance structures of all participating Selves. No single governance authority covers all contributing aspects. The shared substrate operates under governance configurations that are themselves substrate content authored under joint authority (Paper 3 §8). This is the structural-extension move: from one governance authority at intra-Self scope to joint authority at inter-Self scope.

Output. Evolution feed, not offspring. When the shared substrate dissolves at the close of an FAI event, content propagates to *each* participating Self's *home* substrate via governance-configured ingestion at each home perimeter (Paper 3 §7's four-locus mechanism). There are no offspring within a shared parent — the shared substrate dissolves, and what each Self receives feeds its own Paper 2 evolution mechanisms operating at home. The layer-routing rule determines which mechanism receives which content: DNA-layer content feeds DNA evolution at home; action-layer content feeds action-feedback evolution at home; instinct evolution takes no FAI input by architectural commitment.

Three merge patterns. The same three structural patterns apply, with terminology adjusted to the inter-Self register: full merge, governance-configured partial merge, and provenance-carry-over merge. The structural logic is inherited from Paper 2's three variants without modification; only the register shifts.

4. The two properties that distinguish mating from FAI

The disambiguation reduces to two structural differences:

Governance authority scope. Mating (Paper 2) operates under one governance authority — the Self whose lifecycle governance covers all participating entities. FAI (Paper 3) operates under joint governance authority across the governance structures of multiple Selves. This difference is not incidental: it is the structural-extension move that defines what Paper 3 adds at the coordination-primitive level. Joint authority at inter-Self scope is two or more Self-scope authorities operating on a shared substrate object; it produces architectural consequences that single-authority intra-Self mating cannot produce, including governance-configured ingestion asymmetry — different Selves may take different things from the same event under their respective authorities.

Output type. Mating produces offspring — new governed entities that exist within the same Self as the parents, carrying the combined content forward as substrate content under the same governance. FAI produces evolution feed — content that dissolves out of the shared substrate and flows to each participating Self's home substrate via its own evolution mechanisms. This difference follows from the governance authority difference: when no single authority governs a persistent shared offspring, the output must take a form that returns to each Self's authority perimeter rather than residing in a jointly-owned entity.

Both differences are traceable to the same source: the inter-Self scope extension produces joint authority, and joint authority produces evolution-feed outputs rather than jointly-governed offspring.

5. Pattern variant correspondence

The three mating variants and the three FAI merge patterns are the same pattern logic at different scopes. The table below presents the one-to-one correspondence:

Paper 2 mating variant	Paper 3 FAI merge pattern	Shared structural logic
Union	Full merge	All parent/contributed content combined; conflicts preserved as first-class substrate state
Selective merge	Governance-configured partial merge	Governance-directed curation determines content scope before or during combination
Lineage-preserved union	Provenance-carry-over merge	All content combined plus explicit provenance references preserved; carry-over depth configurable

This correspondence confirms that FAI inherits the pattern structure from Paper 2's mating without modifying it. The pattern names differ slightly because the intra-Self register uses lifecycle vocabulary (union, lineage) and the inter-Self register uses coordination vocabulary (merge, provenance carry-over), but the structural logic is the same in each row.

6. Disambiguation rule and prior-art closure

"Mating" or "mating event" without qualification refers to Paper 2's intra-Self lifecycle primitive: two or more entities from the same Self, one governance authority, offspring produced within that Self, governed by the Self's lifecycle governance.

"FAI" refers to Paper 3's inter-Self coordination operation: aspects from different Selves, joint governance authority, evolution feed produced for each participating Self's home substrate, governed by FAI governance under Paper 3's shared-substrate commitments.

When Paper 3 references "mating," it is establishing inheritance — citing the Paper 2 source from which FAI's pattern structure derives — not claiming that FAI is an intra-Self operation or that the two terms are interchangeable.

The prior-art claim this note formalizes: the trilogy establishes two distinct but related coordination primitives — intra-Self mating (Paper 2) and inter-Self FAI (Paper 3) — where FAI is the Paper 2 merge primitive extended to the inter-Self scope, with the extension consisting of joint governance authority and evolution-feed outputs rather than offspring. The two terms name different governance operations; they should not be used interchangeably, and the appearance of "mating" in Paper 3's inheritance accounting does not collapse them into one operation.

Sources

Li, Wenxin. *The Instinct/Reasoning Separation Outside the Model*. Independent publication, April 2026.

Li, Wenxin. *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. Independent publication, April 2026.

Li, Wenxin. *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. Independent publication, April 2026.